

City infrastructure ontologies

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ABSTRACT ORIGINAL

Sustainable urban infrastructure planning and maintenance require an integrated approach that considers various infrastructure assets (e.g., the ground, roads, and buried pipes) and their inter-linkages as a holistic system. To facilitate the usage of this integrated approach, we propose a model of city infrastructure assets and their interdependencies, providing details on how asset properties and processes affect each other. This model is represented as ontologies in OWL 2 Web Ontology Language Manchester Syntax, which can be read and interpreted by machines automatically. These ontologies cover the classifications, properties and processes of the ground, roads and buried water pipes, as well as some related human activities and natural phenomena (e.g., precipitation). The ontologies not only provide a foundation for integrating various types of infrastructure and environmental data, but also for understanding the potential knock-on effects of asset failures. The ontologies have been utilised in a decision support system for integrated urban inter-asset management. © 2023 The Authors